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SUPERSEDING
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MILITARY SPECIFICATION

GENERATOR SET, DIESEL, 750 KW,

50/60 HERTZ, PRIME, UTILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an enclosed, diesel engine-driven, prime, (type II) utility (class 2) generator set, rated for 750 kilowatts (kW) at 60 Hertz (Hz), 625 kW at 50 Hz, 0.8 power factor (pf), lagging.

a. For 60 Hz operation: 4160 volts, 3 phase, 4 wire, wye (2400 volts, line-to-neutral, 4160 volts, line-to-line).
b. For 50 Hz operation: 3800 volts, 3 phase, 4 wire, wye (2200 volts, line-to-neutral, 3800 volts, line-to-line).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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SPECIFICATIONS

FEDERAL

W-B-131	- Battery, Storage Vehicular, Ignition, Lighting, and Starting.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-L-9000	- Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel.
MIL-E-10062	- Engine, Preparation for Shipment and Storage of.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-G-28554	- Generator Sets, Mobile Electric Power and Supplemental Equipment, Packaging of.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engines, Arctic.

STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	- Electromagnetic Interference Characteristics Measurement of.
MIL-STD-705	- Generator Sets, Engine-Driven, Methods of Tests and Instructions.
MIL-STD-889	- Dissimilar metals.
MIL-STD-1332	- Definitions of Tactical, Prime, Precise, and Utility Terminologies for Classification of The DOD Mobile Electric Power Engine.
MIL-STD-1400	- Engine, Gasoline or Diesel, Methods of Test.
MIL-STD-1474	- Noise Limits for Army Materiel.

2.1.2 Other Government drawings. The following other Government drawings form a part of this specification to the extent specified herein.

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DRAWINGS

USAF/SMAMA

TA30554.80-208	- Generator Set, Diesel Eng. 750 KW, 50/60 Hz, Prime, Utility.
TA30554-80-7466	- Remote Control Module.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a 750 kW, 50/60 Hz, 4160 volts, 3 phase, diesel-engine-driven generator set, complete with accessories and auxiliary equipment as specified herein and required for normal operation. The set shall have a removable weathertight housing with engine-generator compartment and control room compartment. The generator set shall be rated for continuous duty operation and shall be capable of a continuous 2-hour overload at 110 percent of the 60 Hz rating from -25° F (-31.7° C) to +125° F (51.7° C) and at altitudes from sea level to 1500 feet (719.8 mmHg). The generator and auxiliaries shall be rated for 4160 volts, 0.8 pf lagging, 60 Hz as an individual set, in parallel with other sets of like characteristics, or with commercial power. The set shall have a remote control station to enable starting, stopping, monitoring operation, synchronizing, and complete control from the control module and a site requirement module for automatic parallel operation up to four sets with a utility or with each other. The set shall be type II (prime), class 2A (utility), mode I (50/60 Hz) as defined in MIL-STD-1332. All items of equipment and components and parts of all generator sets delivered under one contract shall be identified and interchangeable. The generator set shall operate at the following nominal voltage ratings:

a. For 60 Hz operation:

4160 volts, 3 phase, 4 wire, wye (4160 volts line-to-line).

b. For 50 Hz operation:

3800 volts, 3 phase, 4 wire, wye (3800 volts line-to-line).

The generator set shall be a completely housed, skid mounted, factory assembled-and-aligned packaged type.

3.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed drawing requirements is

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permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the set. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.3 Reliability. The specified mean-time-between-failure (MTBF) shall be 1200 hours when tested as specified in 4.8.19.

3.4 First article. The contractor shall furnish two complete assembled sets for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his company can produce sets that comply with the requirements of this specification. Examination and tests shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.4).

3.5 Environmental, operational, and handling. When equipped with all accessories, the set shall start (see 6.3.1), and operate as follows:

3.5.1 Starting.

3.5.1.1 Without engine preheat system. The set shall start within 5 minutes:

a. At ambient temperature from +125° Fahrenheit (F) (51.7° Celsius [C]) to +32° F (0.0° C) at mean sea level (msl) and at any possible relative humidity.
b. With up to 12 inches of rain per hour impinging on the set at angles from the vertical up to 45 degrees from vertical.
c. With the base of the set in planes from level to 5 degrees from level.
d. At any altitude from sea level (760 millimeters of mercury [mmHg]) and a maximum temperature of 125° F (51.7°C) up to and including 8,000 feet (564.4 mmHg) and a maximum temperature of 95° F (35.0° C).

3.5.1.2 With engine preheat system. After the engine preheat system has been in operation for not more than 6 hours, the set shall start and accept not less than 50 percent of rated load within 30 seconds, and 100 percent of rated load within 2 minutes.

a. At ambient temperatures from +32° F (0.0° C) to -25° F (-31.7° C) at sea level conditions and any possible relative humidity.
b. At all other conditions specified in 3.5.1.1.

3.5.2 Storage temperature and humidity. The set shall not be damaged (see 6.3.3), by storage in temperatures from +155° F (68.3° C) to -65° F (-53.9° C) at any possible relative humidity within that range.

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3.5.3 Operating. After the set has started, and after a warmup period as recommended by the engine manufacturer, the set shall operate for a total of 10,000 hours in accordance with the provisions specified herein and under conditions as follows:

a. Without maintenance, overhaul, or replacement of parts, other than lubrication, routine engine servicing, and periodic adjustments as permitted in the schedule of table I.
b. At continuous loads up to and including rated load under the conditions specified in 3.5.1.1, except for the altitude conditions (see 3.5.1.1d).
c. For the altitude condition; the set shall meet the above at rated load at an altitude of 1,500 feet (719.8 mmHg) at +90° F (+32.2° C) not less than 80 percent of rated load at 5,000 feet (632.4 mmHg) at 107° F (+41.7° C), and not less than 75 percent of rated load at 8,000 feet (564.4 mmHg) at +95° F (+35° C).

3.5.4 Operating speeds. The operating (rated) speed of the generator set shall be not greater than 1800 rpm for 60 Hz operation and not greater than 1500 rpm for 50 Hz operation. The set shall operate at speeds up to and including 110 percent of 60 Hz rated speed for a period of not less than 5 minutes without damage.

3.5.5 Flexural vibrations and critical speeds. The set shall be free from dangerous flexural vibration (see 6.3.4) or dangerous torsional vibrations at critical speeds (see 6.3.5), between 80 and 125 percent of rated speed for both 50 Hz and 60 Hz operation (see 3.5.4).

3.5.6 Rough handling. The set shall withstand, without damage (see 6.3.6), rough handling which would normally be encountered during surface or air transporation (see 6.3.7).

TABLE I. Servicing, adjustment, and replacement test schedule.*

No.	Minimum Hours of operation	Servicing, adjustment, and replacement allowed
1	8	Check and add lubricating oil, without stopping engine, as required.
2	8	Add coolant, without stopping engine, as required.
3	8	Check air filter restriction indicator, check fan belts and drain sediment from fuel strainer.
4	100	Check air filter restriction indicator. Change or clean filter panels, when required.
5	250	Change lubricating oil and lubricating oil filters and lubricated accessories or as recommended by the engine manufacturer.
6	250	Replace coolant filter.

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TABLE I. Servicing, adjustment, and replacement test schedule.* (Cont'd)

No.	Minimum Hrs. of operation	Servicing, adjustment, and replacement allowed
7	500	Inspect engine pistons, rings and cylinder liners to the extent possible without disassembly.
8	500	Adjust belt tension.
9	500	Check battery, battery charging system, all terminals and connections. Add water to cells as required.
10	250	Change fuel oil filters. Drain fuel tank condensation.
11	1,500	Check valve tappet clearance and reset, if necessary.
12	4,500	Clean injection nozzles, clean and adjust fuel setting as necessary.
13	2,500	Replace belts as necessary.
14	5,000	Inspect governor.
15	10,000	Replace fuel injectors, as necessary.

* This schedule shall be followed during production inspection (tests) or first article inspection (tests). If maintenance shutdown, as permitted in the schedule, should become due (because of hours accrued during previous testing) after the beginning of the endurance test, the shutdown shall be postponed until the end of the 100 hours interval currently in progress.

3.6 Emergency shutdown. The emergency shutdown located on the engine control panel and the remote control station panel shall stop fuel and air flow to the engine when activated. The shutdown located on the set side shall stop fuel flow only.

3.7 Battery. Unless otherwise specified (see 6.2), 12 volt lead acid storage batteries conforming to W-B-131 and each rated for 200 ampere-hours at the 20-hour rate, shall be furnished and installed. Unless otherwise specified (see 6.2), for production sets, the batteries shall be shipped charged and dry. For first article sets one filling of electrolyte of the proper specific gravity shall be furnished for each battery.

3.7.1 Battery charger. The battery charger shall be in accordance with drawing 80-7935. The battery charger voltmeter and ammeter shall monitor the output voltage and current of the battery charger.

3.8 Sound pressure level. When measured in accordance with 4.8.12, the "A" weighted sound pressure level at 50 feet from the surface of the set shall not exceed 85db, and the "A" weighted sound pressure level inside the control compartment shall not exceed 80db.

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3.9 Emergency lighting system. The battery powered emergency lighting system shall be activated upon loss of station power whenever the emergency lighting switch is "on". The lighting system shall be deactiviated whenever station power has returned or the emergency lighting switch is "off".

3.10 Engine.

3.10.1 Fuels and lubricants. The set shall be capable of satisfactorily starting and operating on fuels conforming to VV-F-800, grades DF-A, DF-1 and DF-2, MIL-F-16884, and turbine fuel conforming to MIL-T-5624, grade JP-5. All components of the engine and fuel system shall be capable of operating on grade JP-5 turbine fuel, except that rated load may be reduced a maximum of 20 percent of rated kW. When operated continuously on grade JP-5 fuel, the set shall meet all requirements specified herein except that the injectors and fuel pumps may be changed and timing adjusted from the diesel setting at the time of changing fuel. The set shall be capable of continuous operation using lubricating oil conforming to MIL-L-9000 and MIL-L-2104 and MIL-L-46167.

3.10.2 Cranking motors limit relays. The cranking limit relay shall defeat the cranking motor circuit and prevent excessive cranking in the event the engine fails to start. Timing of the relay shall be in accordance with drawing 80-7658.

3.10.3 Horsepower. When operating on fuels and lubricants as specified herein, the continuous brake horsepower (bhp) of the engine required to operate the set at rated load, 60 Hz operating speed, shall not exceed 85 percent of the total available horsepower developed.

3.10.4 Governor system performance. When initially adjusted for 0 to 3 percent droop, the governing system shall provide set performance, as specified herein, without adjustments other than permitted (see 3.5.3).

3.10.4.1 Frequency regulation. The frequency regulation for zero droop shall not exceed one-forth of 1 percent of rated frequency for every load change up to and including rated load.

3.10.4.2 Short-term frequency stability (30 seconds). At every constant load from no load to rated load, the speed govering system shall maintain frequency within a bandwidth equal to 0.5 percent of rated frequency. The governing system shall not permit repetitive periodic frequency variation, commonly called hunting, even though within the allowable 0.5 percent band.

3.10.4.3 Long-term frequency stability (4 hours). At constant ambient temperature, constant barometric pressure, constant voltage, and any constant load from no load to rated load, the speed governing system shall maintain frequency within a bandwidth equal to 1.0 percent of rated frequency within a 4-hour operating period.

3.10.4.4 Frequency drift. With the set operating at constant load and voltage, a change in the ambient temperature of 60° F (33.3° C) in an 8-hour

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period shall not cause the frequency to change by more than 1 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.10.4.5 Frequency transient performance. With the speed control system set between 0 and 3 percent operation, and following a sudden 50 percent change in load between no load and rated load, the speed governing system shall reestablish stable engine operating conditions within 4 seconds. The maximum transient frequency change above the new steady state frequency (overshoot) shall be not more than 4 percent of rated frequency. The maximum transient frequency change below the new steady state frequency (undershoot) shall be not more than 4 percent of rated frequency. In addition the circuit breaker shall not trip when the load is suddenly decreased from rated load to no load in one step.

3.10.4.6 Frequency adjustment. It shall be possible to adjust the frequency when the set is operating as follows:

a. At any load up to and including rated load:

   (1) Between 57.5 and 62.5 but not more above 65 Hz for the 60 Hz operating condition.
   (2) Between 47.5 and 52.5 but not below 45 Hz for the 50 Hz operating condition.

b. When the frequency selector switch is in the 60 Hz position the frequency adjusting control shall not allow the operating frequency to be adjusted in the 47.5 and 52.5 Hz range and the frequency adjusting control shall not allow the operating frequency to be adjusted in the 57.5 and 62.5 Hz range when the frequency selector switch is in the 50 Hz position.

3.10.5 Engine protective device. The set output circuit breaker shall open and the engine shall shut down under the following conditions:

a. Coolant temperature exceeds 217° F ±2° F.
b. Oil pressure is less than 10 psig ±2 psi.
c. When the fuel tank is not filled within 35 minutes after the annunciator panel low fuel level light and audible alarm system are activated.
d. The radiator coolant level drops below the lower of the two coolant level switches located on the radiator top tank.
e. The engine speed reaches 2160 ±90 rpm.

3.10.6 Preheat system. The engine preheat system shall maintain the engine in a preheated, ready-to-start condition in ambient temperature from +32° F (0.0° C) to -25° F (31.7° C).

3.10.7 Prelube pump. An engine prelube system shall have sufficient capacity to prelube the engine at any temperature between -25° F and 125° F in one minute to 30 psig pressure with 70° F to 90° F lubricating oil.

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3.11 Indicating instruments. Instruments shall be in accordance with applicable drawings.

3.11.1 Lubricating oil pressure indicator. The lubricating oil pressure indicator shall indicate the pressure of the lubricating oil supply to the engine.

3.11.2 Lubricating oil temperature indicator. The lubricating oil temperature shall indicate the temperature of the oil leaving the engine.

3.11.3 Coolant temperature indicator. The coolant temperature indicator shall indicate the temperature of the coolant leaving the engine.

3.11.4 Fuel level indicator. Fuel level indicator shall show the relative level of fuel in the fuel tank. There shall be no usable fuel remaining when the indicator shows E (corresponding to empty).

3.11.5 Running time meter. The running time meter shall indicate the accumulative hours of engine operation. The meter shall be as specified in drawing 73-0507.

3.11.6 Exhaust temperature indicator. Exhaust temperature indicator shall show the exhaust temperature of the engine left bank and right bank. The indicator shall be as specified in drawing 80-7612.

3.11.7 Watt-hour meter. Watt-hour meter shall show the total accumulated power generated.

3.11.8 Phase sequence indicator. The phase sequence indicator shall be such that, with the selector switch in either the GENERATOR or BUS position, the indicating light on the left will be energized when the phase sequence is 1, 2, 3 (or A, B, C) and the indicating light on the right will be energized when the phase sequence is 3, 2, 1 (or C, B, A). With the selector switch in the OFF position, the phase sequence indicator system shall be deenergized. The "phase sequence lamp test" pushbutton will simultaneously energize both indicating lights with the selector switch in either position.

3.12 Generator.

3.12.1 Generator construction. The generator rotor shall be electrically and mechanically balanced for all speeds up to and including 125 percent of rated speed.

3.12.1.1 Windings and leads. The generator leads shall be sized and braced to withstand the mechanical and electrical stresses resulting from an external fault of up to 250 MVA for the period of time required for operation of the generator protective devices and tripping of the circuit breaker.

3.12.1.2 Dielectric strength. Electric windings shall withstand the following 60 Hz voltages applied for 1 minute:

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a. Generator armature - 9,320 volts between phase windings and 5800 volts between phase windings and ground.
b. Generator field and exciter windings - 10 times ceiling voltage, but neither less than 1,500 nor more than 3,500 volts. (Applied between windings and ground.)
c. Windings energized by the 24 volt dc control, cranking and battery-charging systems - 500 volts. (Applied between windings and ground.)
d. All others - twice rated voltage plus 1,000 volts. (Applied between windings and ground and between windings where applicable.)

3.12.2 Excitation system. The generator excitation system shall consist of the rotating brushless exciter with rectifier assembly, voltage regulator, voltage adjustment control, accessories, and wiring required to provide excitation and shall control the generator output voltage under all operating conditions specified herein. The excitation system shall have adequate capacity to: (a) provide for the specified set performances; (b) cause the set output voltage to rise to not less than 125 percent of rated value at no load, hot field, rated frequency conditions, at 125° F (51.7° C); (c) support the specified minimum short circuit; and (d) provide sufficient current to initiate operation of current sensitive protective devices.

3.12.3 Exciter field current. The exciter field current of production generators shall not differ by more than 10 percent from the field current required by the first article generators under the same conditions.

3.12.4 Winding resistance. The winding resistance shall be in accordance with the applicable drawings. All resistance values shall be corrected to winding temperature of 25° C (77° F).

3.12.5 Generator performance. The generator, together with the excitation system, shall provide performance as specified herein. The generator shall have an inherent limit of 160 percent of rated voltage for any condition of load and field at rated frequency and any ambient condition specified herein.

3.12.5.1 Voltage waveform. The square root of the sum of the squares of the harmonic amplitudes for the line-to-line and line-to neutral voltage, for each voltage connection, shall be not more than 5 percent, nor shall any single voltage harmonic be more than 2 percent under the following output conditions of the generator, when under automatic control of the excitation system:

a. No load.
b. At any load up to and including rated kW and at any pf between unity and 0.8 lagging.

3.12.5.2 Voltage unbalance. With the generator under control of the excitation systems, and operating at no load with rated voltage and frequency, the maximum difference in the three line-to-neutral voltages shall be not more than 1 percent of the rated line-to-neutral voltage. The voltage unbalance between phases shall not exceed 5 percent of rated voltage under the following simultaneous load conditions:

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a. The generator connected for 3 phase output.
b. A single phase, unity pf load connected line-to-line, with no other load on the set.
c. A load current of 25 percent of rated load current of the set.

3.12.5.3 Voltage regulation. The voltage regulation (see 6.3.12) of the generator set shall be not more than 2 percent of rated voltage.

3.12.5.4 Voltage-short term stability (30 seconds). At every constant load from no load to rated load, the voltage at the set terminals shall remain a bandwidth equal to 1 percent of rated voltage.

3.12.5.5 Voltage-long term stability (4 hours). At constant ambient temperature, constant barometric pressure, constant frequency, and any constant load from no load to rated load, the voltage shall remain within a bandwidth of 2 percent of rated voltage in a 4 hour operating period.

3.12.5.6 Voltage drift. With the set operating at constant load and frequency, and change in ambient temperature up to 60° F (33.3° C) in an 8 hour period shall not cause the voltage to change by more than 1 percent, set temperature stabilization being accomplished at both the initial and final ambient temperature conditions.

3.12.5.7 Voltage-transient performance. With the reactive load compensation of the voltage regulator inoperative, and with the engine speed control system set for 0 to 3 percent regulation, performance of the set under transient conditions as measured by an oscillograph, shall be as follows:

a. With the set initially operating at rated frequency and rated voltage, and following any sudden change in load from no load to half load, and half load to full load, the instantaneous root mean square voltage shall not drop to less than 80 percent of rated voltage and shall reach stable conditions within 3 seconds. No overshoot or undershoot of the final voltage may exceed the initial voltage transient in amplitude.
b. The above requirements shall also apply when load is suddenly changed from half load to no load, except that the initial voltage transient shall involve a voltage rise not to exceed 130 percent of rated voltage.
c. With the set initially operating at no load, rated voltage and rated frequency, the instantaneous root mean square voltage shall not drop to less than 60 percent of the no load rated voltage when a balanced 3-phase, 0.4 pf or less, lagging static load having an impedance of 0.5 per unit is applied to the output terminals of the set. The generator shall recover to not less than 95 percent of rated voltage within 5 seconds and shall stabilize at or above this voltage.

3.12.5.8 Voltage operating range. The output voltage of the set shall be adjustable to any value within the following line-to-line voltage operating range:

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a. At 60 Hz: Between 3,750 and 4,575 volts.
b. At 50 Hz: Between 3,420 and 4,180 volts.

The set shall operate satisfactorily at any load between no load and rated load throughout the voltage operating range.

3.12.5.9 Short circuit. The generator, exciter, and voltage regulator operating as a unit shall withstand the following short circuits at its output terminals, when operating at rated load, without reduction of the dielectric strength to a point where it will not meet the requirements of 3.12.1.2:

a. A 10-second symmetrical three-phase short circuit with a sustained short circuit current of not less than 300 percent of rated output current.
b. Line-to-line short circuits of sufficient duration to produce an I^2T value of not less than 160, using the current at time T. The current at time T shall not be less than 300 percent of rated output current.
c. Line to neutral short circuits of sufficient duration to produce an I^2T value of not less than 200, using the current at time T. The current at time T shall not be less than 300 percent of rated output current.

3.12.5.10 Parallel operation. The set shall be capable of parallel operation at zero speed droop with not more than 2 percent voltage droop and at 3 percent speed droop with not more than 3 percent voltage droop. The governor parallel control and reactive load compensation control shall remain in the same position during both conditions.

3.12.5.10.1 Parallel operation with zero speed droop. With their governor and voltage regulator paralleling circuits properly interconnected, any number of like sets operated in parallel shall divide load in accordance with the following, as system load at rated power factor is varied between 20 and 100 percent (and vice versa) of the combined rating of the connected sets:

a. Real power division. The difference between the average individual kW output shall at no time be greater than 10 percent of the kW rating of one set.
b. Power exchange. At any constant system load, power exchange between the sets shall not exceed 10 percent of the rating of one set. (Power exchange is the difference between the maximum and minimum power output of one set, for constant system load conditions, as determined from oscillographic measurements.)
c. The reactive power division. The difference between the average reactive kilovolt amperes (kvar) supplied shall at no time be greater than 10 percent of the reactive kvar rating of one set.

Therefore, no adjustment shall be made to the governors, voltage regulators, or other components of the set as the system load is varied. For the parallel operation test, the initial adjustments to ensure that both KW and KVAR are equally divided between sets shall be made with a system load of 25 percent of the system rating.

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configuration shall be consistent with requirements of

3.12.5.10.2 Parallel operation with speed droop. With the governor droop switch in the droop position and the voltage regulator shorting plug installed in the paralleling receptacle, like sets operated in parallel shall divide load in accordance with 3.12.5.10.1, as system load at rated power factor is varied between 0 and 100 percent (and vice versa) of the combined rating of the connected sets.

3.12.6 Generator protective devices. Operation of these protective devices shall trip the generator circuit breaker, sound the alarm, and indicate the respective operation on the annunciator alarm system.

3.12.6.1 Overvoltage. The overvoltage protective device shall actuate at 239 ±5 volts in 3 ±1 second. This device shall also simultaneously stop the engine.

3.12.6.2 Undervoltage. The undervoltage protective device shall actuate at 166 volts ±5 volts in 1.5 ±1 second.

3.12.6.3 Overcurrent. The overcurrent protective device shall operate when reactive load on all three phases reaches 130 percent (169 amp) of the rated current; the device shall trip in 20 ±5 seconds.

3.12.6.4 Synchronizing check relay. With the parallel switch in the parallel position and the synchronizing switch in the on position, the synchronizing check relay shall verify that the generator is synchronized (proper voltage, frequency, and phase coincidence) with the bus before the circuit breaker is closed.

3.12.6.5 Reverse power. The reverse power protection device shall actuate when the 3 phase reverse power reaches and remains at 20 percent ±3 percent for 0.2 seconds.

3.12.6.6 Ground fault. The ground fault relay shall actuate in 0.5 to 0.9 seconds.

3.12.6.7 Underfrequency. The underfrequency protective device shall activate when the output frequency drops below 46 Hz ±1 Hz for 50 Hz operation and 56 Hz ±1 Hz for 60 Hz operation. The device shall not permit closure of the output circuit breaker at frequencies below those specified above.

3.13 Annunciator alarm system. When a monitoring or protection device operates, the annunciator alarm system shall indicate the condition by sounding the audible alarm and by a flashing light behind the respective window(s) of the annunciator panel. Operation of the "ALARM SILENCE" pushbutton shall silence the audible alarm and cause the light(s) to stop flashing, but remain lighted. After the cause of alarm has been corrected, operation of the "RESET" pushbutton shall extinguish the light(s) and reset the system. Operation of the "TEST" pushbutton shall sound the audible alarm and light all windows of the annunciator panel as long as the pushbutton is held in the depressed position. The following conditions shall activate the annunciator alarm system:

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a. "HIGH COOLANT TEMPERATURE": Coolant temperature exceeds 208° F ±4° F.
b. "HIGH LUBRICATING OIL TEMPERATURE": Lube oil temperature exceeds 245° F ±3° F.
c. "LOW LUBRICATING OIL PRESSURE": Lube oil pressure is less than 39 psig ±2 psi.
d. "LOW FUEL LEVEL": Fuel level in tank is sufficient for only 34 to 36 minutes of operation at rated load.
e. "HIGH FUEL LEVEL": Fuel level prior to fuel overflowing.
f. "OVERSPEED": (See 3.10.5).
g. "OVERVOLTAGE": (See drawing 80-7794).
h. "UNDERVOLTAGE": (See drawing 80-7795).
i. "OVERCURRENT": (See drawing 80-7395).
j. "REVERSE POWER": (See drawing 80-7797).
k. "GROUND FAULT": (See drawing 80-7436).
l. "UNDERFREQUENCY": (See drawing 80-7380).
m. "LOW COOLANT LEVEL": Coolant level drops below the higher of the two coolant level switches in the radiator.
n. "EXCESSIVE CRANKING": (See drawing 80-7658).

3.14 Switch gear. The generator output switchgear shall be in accordance with the applicable drawings and as specified herein.

3.14.1 Output circuit breaker. When paralleling the generator with other like power sources, the closing circuit of the circuit breaker shall be electrically interlocked with the parallel switch, the synchronizing switch, and the synchronizing check relay such that the following conditions must be met before the circuit breakers can be closed:

a. Parallel switch is in the PARALLEL position.
b. Synchronizing switch is in the ON position.
c. Synchronizing check relay has verified that the generator is synchronized (proper voltage, frequency, and phase) with the bus (load side of circuit breaker).

3.14.2 Disconnect switch. The disconnect switch shall be interlocked with the output circuit breaker so that the circuit breaker must be in the "OPEN" position before the disconnect switch can be opened and so that the disconnect switch must be "CLOSED" before the circuit breaker can be closed.

3.14.3 Maintenance lockout switch. All circuits except the convenience receptacle and the lights shall be de-energized when the maintenance lockout switch is in the "MAINTENANCE" position.

3.14.4 Parallel switch. When the parallel switch is momentarily placed in the "PARALLEL" position, the reactive load compensation control shall be fully operative and the synchronizing switch, the synchronizing check relay, and the closing circuit of the circuit breaker shall be electrically interlocked (see 3.14.5). When the switch is momentairly placed in the "SINGLE" position, the reactive load compensation control shall be inoperative and the interlock between the synchronizing switch, the synchronizing check relay, and the closing circuit of the circuit breaker shall be bypassed.

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3.14.5 Synchronizing switch. The synchronizing switch, when in the "ON" position, shall energize the synchroscope and synchronizing lights. The synchronizing switch shall be electrically interlocked with the parallel switch, the synchronizing check relay, and the closing circuit of the circuit breaker so that, with the parallel switch in the PARALLEL position, the synchronizing switch must be in the ON position and the synchronizing check relay must verify that the generator is synchronized with the bus before the circuit breaker can be closed.

3.15 External station power source. External station power shall be either 120/240 volts single phase 3 wire or 380 volts single phase. The external station power circuit breaker shall be interlocked with the set station power circuit breaker to prevent both sources from being connected simultaneously to the station power bus.

3.16 Fuel transfer system. With station power available and the fuel transfer selector switch in the "AUTO" position, the fuel transfer system shall be controlled by the float level switch in the fuel tank to maintain sufficient fuel in the fuel tank for continuous operation at rated load.

3.17 Remote control. When specified (see 6.2), each generator set shall be furnished with a separate remote control station and remote control cable. When used, the remote control station and cable will allow remote start, stop, monitor, control of the set in unit operation, and paralleling with like sets, from a location up to 100 feet away. Reference TA30554-80-7466.

3.18 Site requirement module (SRM). When automatic operation is specified (see 6.2) a site requirement module (SRM) shall be provided in accordance with drawing 80-7200 and 80-7300 as specified herein. The SRM shall operate as follows:

a. When the "Peaking On" switch is depressed, the generator sets shall start and synchronize with the utility. When the sets are on-line, the generator load control shall ramp load each set from zero to rated load. To stop the sets, depress the "Peaking Off" switch. The generator load control shall ramp unload each set from rated load to no load. The set circuit breaker shall be opened to allow the set(s) to run unloaded, for five minutes to cool down. The set(s) will then be shut off. When in the "Normal Standby" mode, a failure of the utility shall cause the utility circuit breaker to lock open. All the load breakers shall then be opened. The load breakers shall have a predetermined order of priority and are pre-selected so as not to exceed the rating of any one set. A failure of the utility is defined as when the voltage varies by more than ±10 percent and the frequency varies by more than ±3.0 percent for 10 seconds.
b. All sets shall start.
c. The first generator to reach the proper voltage and frequency shall close to the bus.
d. The first priority load breaker shall close.
e. The remaining sets shall synchronize with the bus and close their breakers.

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f. A pair of contacts per on-line set shall be provided for load breaker control. The first contact shall close when a generator set breaker closes. The second contact shall close approximately three seconds after the first contact is closed. These two contacts shall cause step loading assuming that no step exceeds 50 percent rating of each set. These contact pairs shall be prioritized such that each contact pair will close in order from highest priority to lowest priority as each generator set closes to the bus. The order in which the generator sets close to the bus shall not affect the closing order of the contact pairs. A shutdown of any set shall open the least priority load breaker.

g. Upon the return of the utility, all sets shall be synchronized to the utility, by a master synchronizer. When all sets are synchronized to the utility, the utility circuit breaker shall close. The generator load control and the KVAR controller will be activated immediately. The generator load control shall be at the maximum setting when synchronization is achieved. Cross current loop shall be open if necessary when the utility breaker is closed. The KVAR network shall be activated to control the system voltage as follows:

(1) In the automatic mode, a fixed resistor shall be connected to the circuit, in place of the adjustable resistor. This fixed resistor shall insure that all generators run at the same voltage. The internal range adjustment of each regulator shall be adjusted to compensate for tolerance of the resistor and the system dynamics.

(2) The Motor Operated Control (MOC) receives error signals from either of two sources:

(a) Prior to synchronizing with the utility, the MOC would be controlled by the automatic synchronizer.

(b) After synchronization, the KVAR controller shall be activated and shall now control the MOC to control the VARS.

(3) The SRM shall have a current transformer for the KVAR controller.

(4) The KVAR controller shall be programmable to respond to the number of sets on-line.

h. The sets shall be ramp unloaded, the breakers opended and the sets run unloaded for five minutes to cool down. The sets shall then be shutdown.

i. The time required to parallel each set to the bus after the set has reached rated voltage and frequency shall not exceed five seconds. The set shall be considered paralleled to the bus when the set has synchronized to the bus and closed the output circuit breaker.

3.19 Interference. The electromagnetic interference emission and susceptibility characteristics of the set shall not exceed the UMO4 limits for class C2 equipment of MIL-STD-461 even though the set KVA rating is greater than that specified in the standard.

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3.20 Air transportation. The set shall be suitable for transport as a palletized load on a C-130, C-141 and C-5A aircraft.

3.21 Workmanship. Workmanship shall be in accordance with requirement 9 Of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government reserves the right to reject any equipment for not meeting any requirement of this specification even though not performing a test directly related to the specific requirement.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.6).
c. Inspection of packaging (see 4.9).
d. Inspection comparison (see 4.4).

4.3 First article inspection. First article set(s) shall be examined and tested as specified herein. The inspection reports and inspection records of the generator set shall describe:

a. Malfunction, evidence of damage, failure, or adjustment, other than adjustments permitted in this specification and the applicable test methods, which occur during the examination and tests.
b. The cause of the malfunction, damage, or failure and reason for the adjustment.
c. The corrective action taken or required.

A report covering inspection of the first article set shall be forwarded to the contracting officer upon completion of inspection.

4.3.1 Examination. Each set shall be examined, without disassembly for conformance to the drawings (see 3.2). Nonconformance with the drawings shall be cause for rejection.

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4.3.2 Tests. Performance and test results which show that the generator or generator sets do not meet the requirements of this specification shall be cause for rejection of the equipment.

4.3.2.1 Generators. The generators shall be subjected to all tests marked "X" under column 1 of table II.

4.3.2.2 Generator sets. The sets shall be subjected to all the tests marked "X" under column 2 of table II after the first article generators have passed all tests marked "X" listed under column 1 of table II. The tests marked "X" under column 2 of table II may be conducted in any order, except the last test shall be reliability.

4.3.2.3 Table III, schedule of optional tests. The optional tests as indicated in table III shall be conducted by the contractor to determine set compliance subsequent to changes in parts or material when changes are determined major changes by the contracting officer. All quality assurance provisions of section 4 shall apply.

TABLE II. Test schedule.*

First Article		Quality Conformance		
Generators	Generator Sets	Individual Gen. Sets	Test	Test Method (MIL-STD-705) or Test Paragraph
1	2	3	4	5
X	-	X	Insulation-resistance measurements.	301.1 (see Note 1)
X	-	X	High potential.	302.1 (see Note 1)
X	-	X	Winding-resistance measurements.	401.1 (measurement stator and field windings)
-	X	-	Railroad impact.	740.5 (see Note 4)
X	-	-	Open circuit saturation curve.	410.1 (at both 50 Hz and 60 Hz)
X	-	-	Short circuit saturation curve.	411.1 (at both 50 Hz and 60 Hz)
X	-	-	Zero power factor.	412.1 (at both
X	-	-	Rated load saturation curve.	413.1 (at both 50 Hz and 60 Hz)
X	-	-	Direct-axis transient reactance.	425.1
X	-	-	Negative sequence reactance and impedance.	422.1
X	-	-	Short circuit (mechanical strength).	4.8.9
-	X	X	Frequency adjustment range.	511.2 (at both 50 and 60 Hz)

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TABLE II. Test schedule.* (Cont'd)

First Article		Quality Conformance		
Generators	Generator Sets	Individual Gen. Sets	Test	Test Method (MIL-STD-705) or Test Paragraph
1	2	3	4	5
-	X	-	Voltage unbalance with unbalanced load.	620.2
-	X	-	Fuel consumption.	670.1 (see 4.7.2)
-	X	-	Starting and operating.	4.8.13 at -25° F (31.7° C). Use preheat system for 6 hours.
-	X	-	Voltage dip for zero power factor load.	619.1
-	X	-	Voltage dip and rise for rated load.	619.2 (at both 0-50% and 50-100% rated load conditions).
-	X	X	Parallel operation.	630.1 (at both 50 and 60 Hz operation. All production sets shall be paralleled with the preproduction set in accordance with 4.8.4).
-	X	X	Start and stop.	503.1
-	X	X	Regulator and governor stability and transient	608.1 and 608.2 (both at 50 and 60 Hz).
X	-	-	Overspeed (generator).	505.3 (at 125% rated 60 Hz speed for 15 minutes.)
-	X	X	Overspeed (set).	505.1 (at 110% rated 60 Hz speed for 5 minutes.)
-	X	X	Overspeed protection devices.	505.2
-	X	X	Phase balance (voltage).	508.1 (to be measured at load terminals.)
-	X	-	Circuit breaker (short circuit).	512.1 (use oscillograph)

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TABLE II. Test schedule.* (Cont'd)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test Paragraph
Generators	Generator Sets	Individual Gen. Sets		
1	2	3	4	5
-	-	X	Circuit breaker (short circuit)	5.12.1 (L_1-L_0, L_2-L_0, L_3-L_0 shorts only, oscillograph need not be used.)
-	X	X	Circuit breaker (overload trip).	512.2
-	X	X	Circuit breaker (undervoltage and overvoltage trip).	512.3
-	X	X	Instrument check.	513.1
-	X	X	Under frequency trip.	506.1 (at both 50 and 60 Hz)
-	X	X	Low-oil pressure protective device.	515.1
-	X	-	High temperature	710.1 (Take temperature rise measurements on the rotor (see Note 6) and stator windings under the following conditions at 125° F (51.7° C): 50 Hz, 3800 volts, 2200/3800, 60 Hz, 4160 volts, 2400/4160. Also perform methods 619.1 and 619.2 2400/4160 60 Hz condtion only.
-	X	X	Remote control and operation.	See 4.8.6
-	-	X	Maximum power.	640.4 (see Note 5)
	X	X	Low coolant level.	4.8.15
	X	X	Preheat test.	4.8.18 (see 3.10.6)
-	-	X	Fuel transportation.	4.8.16

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TABLE II. Test schedule.* (Cont'd)

First Article		Quality Conformance	Test	Test Method (MIL-STD-705) or Test Paragraph
Generators	Generator Sets	Individual Gen. Sets		
1	2	3	4	5
-	-	X	Lighting system.	4.8.16
-	-	X	Personnel ventilation systems.	4.8.16
-	-	X	Heating system. a. Personnel. b. Switchgear. c. Generator. d. Lube oil. e. Coolant.	4.8.16 (Note 8)
-	-	X	Prelubrication system.	Section 4
-	-	X	Battery charging system.	4.8.16 (Note 9)
-	X	-	Battery charging system.	4.8.16 (Note 10)
-	-	X	Controls. a. Maintenance lockout switch. b. Parallel switch. c. Emergency shutdown. d. Station power circuit breaker. e. Synchronizing switch.	4.8.16 (see Note 7)
-	-	X	Instrumentation. a. Generator temperature. b. Engine exhaust temperature. c. Clocks. d. Watt-hour meter. e. Running time meter. f. Coolant temperature. g. Lube oil temperature. h. Lube oil pressure. i. Fuel level indicator. j. Phase sequence.	4.8.16

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TABLE II. Test schedule.* (Cont'd)

First Article		Quality Conformance		
Generators	Generator Sets	Individual Gen. Sets	Test	Test Method (MIL-STD-705) or Test Paragraph
1	2	3	4	5
			k. Battery charger ammeter. l. Battery charger voltmeter.	
-	X	X	Annunciator alarm system.	4.8.5
-	X	-	Overload.	4.8.1 at 125° F (51.7° C)

* Perform test marked "X"

TABLE III. Schedule of optional tests.

First Article		Quality Conformance		
Generators	Generator Set	Individual Gen. Sets	Test	Test Method (MIL-STD-705) or Test Paragraph
1	2	3	4	5
-	X	-	Air transportability.	4.8.8
-	X	X	Voltage waveform.	601.4 (At all rated voltage and frequency connections and the following loading conditions.) See Note 2: a. No load. b. Rated load at 0.8 pf lagging. c. Rated load at 1.0 pf.
-	X	-	Humidity.	711.1

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TABLE III. Schedule of optional tests. (Cont'd)

First Article		Quality Conformance		
Generators	Generator Set	Individual Gen. Sets	Test	Test Method (MIL-STD-705) or Test Paragraph
1	2	3	4	5
-	X	X	Site requirement module.	4.8.11
-	X	-	Tiedown.	4.8.3
-	X	-	Lifting and towing.	4.8.14
-	X	-	Electromagnetic interference.	At all rated voltage and frequency connections add the following load conditions: a. No load. b. Rated load at 0.8 pf lagging. c. Rated load at 1.0 pf. Also see 4.8.10 and Appendix A.
-	X	-	Rain.	3.5.1
-	X	-	Torsiongraphing.	504.2
-	X	-	Sound level.	4.8.12 (See Note 3)
-	X	-	Reliability endurance test.	4.8.19
-	X	-	Inclined operation.	660.1 (5 degrees only)
-	X	-	Voltage and frequency drift.	4.8.7
-	X	X	Voltage modulation.	602.1
-	X	-	Horsepower.	MIL-STD-1400 Method 1311

NOTES:

1. Insulation resistance and high potential tests may be made after assembly of the engine to the generator but before final assembly of the complete set. On production sets, the insulation resistance and winding resistance may be performed by the generator manufacturer.

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2. Also, view waveform with oscilloscope. For individual tests, only method 601.4 need be used. The values obtained by test method 601.4 shall be added together and compared to the requirement of 3.12.5.1.

3. Exterior readings for sound level tests are to be taken at not less than six positions, one at each end and two at each side of the set. The interior readings for sound level tests are to be taken at not less than three positions located 2 feet from the front of the control center and not less than four positions in other work areas that are representative of required operator positions when the set is operating. Readings shall be taken at rated load and at no load with the set operating at rated voltage and frequency. The microphone height shall be 5 feet 8 inches.

4. The set shall be humped at 5 mph only. The set shall be wet with all liquids at operating level. One set only shall be tested; once with the generator end forward and once with the control room forward, for a total of 2 impacts in each direction at 5 mph. The set shall be secured to the railroad car using the tiedown provisions provided by drawing requirements. All reference to generator in test method 740.5 shall be changed to generator set.

5. Maximum power of production sets shall not be less than 95 percent of the average value of the first article sets maximum power prior to endurance test.

6. Test data submitted by generator manufacturer will be an acceptable substitute.

7. To avoid stressing the engine, proper operation of the airboxes may be demonstrated with the engine not running.

8. This test is to verify that all heating systems will operate at their electrical power rating as measured at the input terminals of the heating system.

9. This test is to demonstrate the battery charger ability to charge a partially discharged battery to the proper float voltage in accordance with the requirements.

10. Check ability to vary charging current (and voltage) with the voltage adjusting device; check action of current limiter to limit charging current to battery capacity (use discharged battery and adjust for highest possible output voltage); check for effect of alternator output voltage changes or charging current; observe charging ammeter during other tests required herein and note any evidence of nonconformance to requirements.

4.4 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to any examination and test specified herein, necessary to determine that the selected sets meet all

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requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Sets will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article sets. Acceptance of an inspection comparison set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in section 4. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5 Inspection failure. Failure of a set to meet any requirement specified herein during and as a result of the examination and test shall be cause for rejection of the set and shall be cause for refusal by the Government to continue acceptance of sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of such testing will be considered prima facie evidence that all sets accepted prior to the completion of this testing are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the Government.

4.6 Quality conformance inspection.

4.6.1 Examination. Each set shall be examined for conformance to the drawings and compared with the first article set. Each set which does not meet drawing requirements or which deviates from the first article sets shall be considered to be defective.

4.6.2 Tests of each set. Each set shall be subjected to the tests marked "x" under column 3 of table II in the order listed. A set which fails to pass a test shall be rejected.

4.7 Inspection schedule. Table II and the drawings shall be used for first article and quality conformance inspection as specified in 4.3 and 4.6. Examination for conformance to the drawings shall determine compliance with this specification.

4.8 Test procedures. Tests shall be conducted in accordance with MIL-STD-705 and as specified herein. During all tests, except as otherwise specified in table II, the set shall use fuel and lubricants as specified (see 3.10.1). Test instruments shall be of the laboratory type and shall have been calibrated within 30 days of the start of testing and at 6-month periods thereafter. Instruments used in calibration should have at least five times the accuracy of the instrument being calibrated. Calibrated reference instruments of lesser accuracy than standard and which are not used for any other purpose may be used for the required periodic check of test instruments. Instruments shall be calibrated at the frequencies at which they are going to be used. Direct-reading

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instruments shall have at least 0.5 percent accuracy and shall be connected to indicate in the most accurate portion of their range. On dc instruments, the readings shall not be made on the lower 15 percent of the scale. On ac instruments, the readings shall not be made on the lower one-third of the scale. When the test methods call for the use of Esterline-Angus recording type meters, the Texas Instrument Company model PDRHXFHXVA-A16-XT shall be used. Oscillograph galvanometer frequency response shall be flat from "dc" to not less than 3,000 cycles per second. When recording meters are specified for any part of a test, turn on the recording meters prior to starting the warm-up period of the set and record continuously for the duration of the test. The recording meter shall be operated at a minimum speed of 6 inches per hour during the portions of the test where steady state loading conditions exist and shall be operated at a minimum speed of 6 inches per minute at least 30 seconds before, during, and after a load change. All tests shall be conducted with the generator connected for 2400/4160 volts, 60 Hz except as otherwise specified in table II. Reactive load banks shall not saturate at the voltage being used. The batteries used for testing shall be like those supplied with the set. When test method 608.1, MIL-STD-705 is performed, load application steps shall be 0-1/2, 1/4-3/4, and 1/2-4/4, with each load being applied and dropped four times. When test method requires no load to full load changes, the no load to full load change shall be made by changing from no load to 1/2 load allowing 30 seconds for stabilization changing from 1/2 load to full load.

4.8.1 Overload test. Operate the set at rated load and frequency until the set is stabilized. Increase the load in one step to 110 percent of rated 60 Hz load and operate for two hours. Record radiator coolant and oil temperatures. Reduce load to rated load and operate for 30 minutes.

4.8.2 Fuel consumption test. The fuel consumption test shall be conducted at 1/4, 1/2, 3/4, 4/4 and 110 percent of rated load. Results shall be provided in pounds per kilowatt-hour and gallons per hours. The fuel consumption test time may be reduced to two hours at 110 percent load. The fuel consumption test may be conducted as part of endurance run - except at 110 percent load. Applicable charts and data sheets shall be annotated to indicate this test period.

4.8.3 Lifting and tiedown. In lieu of the test, an analytical proof and tiedown diagram may be substituted to show that the equipment will conform to the drawing requirements.

4.8.4 Parallel operation test for production sets. Each production set shall be tested in parallel with the first article set(s). This test shall determine proper operation of all parallel operation controls and devices including the reverse power safety devices. It shall also check for proper load division and kilovolt amperes division at no load, half load, and full load. Set instrumentation may be utilized.

4.8.5 Annunciator alarm system. Test annunciator alarm system for correct operation and indication of all malfunctions (see 3.13).

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4.8.6 Remote control and operation test. The remote control and operation test shall be conducted with the set and remote control station interconnected by use of the 100 foot cable specified. By remote operation, determine compliance of: engine start and stop; voltage and frequency adjustment; application and removal of load; provision to monitor set and bus frequency and voltages, real and reactive power, annunciation system; and parallel operation.

4.8.7 Voltage and frequency drift test. The voltage and frequency drift test shall be conducted immediately following the -25° F test. Operate the set at rated load at -25° F until set is stabilized. Within 8 hours increase the ambient temperature 60° F. The increase in temperature shall not be less than 5° F (2.8° C) per hour. Operate at the new temperature until the set has stabilized.

4.8.8 Air transportability test. One generator set shall be transported to a site specified by the USAF. The air transportability loading will be performed by USAF personnel. The number of aircraft to be test loaded will be determined at the time of the Air Force review of the specific hardware developed.

4.8.9 Short circuit (mechanical strength). Conduct test connections, at both 50 and 60 Hz in accordance with test method 625.1C, as specified herein:

a. Reference 625.1.3.2(a), line 1, delete reference (two places) to "set" and substitute "generator and excitation system".
b. Reference 625.1.3.2(c), delete in its entirety and substitute "repeat steps (a) and (b) above" at all phase line-to-line (L_1-L_2; L_2-L_3; L_3-L_1) single phase line-to-neutral, wye only, line (L_1-L_0; L_2-L_0; L_3-L_0) and symmetrical 3-phase short circuits.
c. In lieu of the stabilization procedure of 625.1.3.2(a). The following procedure may be used:

    1. INITIAL STABILIZATION. Generator and exciter system shall be stabilized at rated load, rated voltage and rated frequency as defined in 625.1.3.2(a). Thermocouples on generator stator winding will also be monitored on a recorder.
    2. SUBSEQUENT STABILIZATIONS. Short circuits will be applied; after each short circuit, the generator will be allowed to cool until the thermocouples return to within approximately 5° C of the recorded temperature after stabilization under (1), before the next short circuit is applied.

4.8.10 Electromagnetic interference test. The set shall be evaluated in accordance with MIL-STD-462 and the supplementary information contained in Appendix A.

4.8.11 Site requirement module (SRM) test. The SRM shall initiate the unattended sequence when the normal 120 volt ac supply voltage varies more than ±10 percent and the frequency varies ±3.0 percent. The system shall consist of two generator sets, one site requirement module and utility power. The utility

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power may be simulated by another 750 kW generator set. The items shall be interconnected to provide for automatic operation. Set instrumentation may be used to determine load on the set. The SRM shall be tested as follows:

a. The sets shall be preheated for six hours, the speed droop set at zero and not more than two percent voltage regulation. The load shall be 1500 kW, 0.8 pf.
b. Adjust the 120 volts to the automatic start circuits to 110 volts and to 130 volts. The set shall not start.
c. Reduce the 120 volts to the automatic start circuits to 107 volts and determine that all sets (1) start, (2) reach the ready to load condition, (3) act to disable the utility breaker, (4) determine that the critical load circuit breaker contacts 52c1 close and the 52f1 contacts close, and the sets have paralleled.
d. Determine that pickup circuits close within the time limit.
e. Determine which one of the sets was selected as the master set.
f. Increase the load of the sets in parallel in approximately 300 kW steps until a 1500 kW total load is reached and operate for one hour.
g. Reduce the load to zero in approximately 300 kW steps and stop the sets.
h. Adjust the frequency to the automatic start circuits to 58.5 Hz and 61.5 Hz. The sets shall not start.
i. By reducing the frequency to the automatic start circuits to 57.5 Hz, determine that the sets do not start until the preselected time delay has elapsed and that the sets start and the critical load and feeder breaker contacts close. If the same set was selected as the master, stop the sets and restart the sets of INSURING that a new master is selected.
j. Increase the load to 1500 kW in approximately 300 kW steps and operate for one hour.
k. Reduce the load in approximately 300 kW steps and stop the sets.
l. Repeat steps "a" through "k" with the system in the 50 Hz configuration with proper derating for load and adjustment for frequency.
m. Initially separate the sets by 3 percent in frequency. Shut the sets down and initiate automatic start and parallel at no load. Apply rated load (1500 kW) within 20 seconds.
n. Repeat "m" with the voltage initially separated by 10 percent.
o. Initially separate the sets by 3 percent frequency and 10 percent voltage. Shut the sets down. Simulate the automatic pick-up of 1500 kW from the utility bus with automatic initiation of load transfer from the bus. The sets shall start, parallel and accept the load from the utility bus and open the tie breaker.
p. With the sets paralleled and carrying 1500 kW, simulate an automatic load transfer from the sets back to the utility bus and shut down the sets.
q. **Contractor shall develop and provide a test plan for incorporating all action "a" thru "p" above and incorporating a test of the requirements of the site requirements module (see 3.18). Test plan shall be submitted to the contracting officer for approval.**

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4.8.12 Sound level test. The sound level test shall be conducted in accordance with 5.1 test requirements of MIL-STD-1474.

4.8.13 Starting and operating, moderate cold. Conduct in accordance with test method 701.2, as modified herein:

a. Paragraph 701.2.1, second line - Delete "without the use of winterization equipment" and substitute "with use of the engine preheat system for 6 hours."
b. Paragraph 701.2.3.2 - Page 4 - Delete subparagraph (e), (f), (g), and (h) and substitute "(e) By following the instruction on the set or in the technical manual start the generator set. Record the time when cranking started. Start shall be initiated by simulating a power failure. (f) Within 30 seconds after initiating the start apply a load equal to not less than 50 percent of rated load. (g) Within 120 seconds after initiating the start the set shall accept not less than 100 percent of rated load. (h) Leave rated load on the set for a minimum of 30 seconds, then drop the load to no load in one step using the circuit interrupter. Operate at no load for a minimum of 30 seconds. Again, using the circuit interrupter, apply and drop load steps two more times with a minimum of 30 seconds of operation in each load condition. Next apply rated load and operate for 5 minutes."

4.8.14 Lifting and towing. Analytical proof shall be provided to show that the equipment will conform to the requirements of the applicable drawings.

4.8.15 Low coolant level test. With the set operating at rated speed, no load and with the circuit breaker closed, open the coolant drain and record the level of coolant when the warning alarm sounds and when the circuit breaker opens and the set stops.

4.8.16 Functional test. For each mode of operation, demonstrate that the specified system/component functions in accordance with the applicable requirements. These tests may be included as a part of other tests.

4.8.17 Humidity. Conduct in accordance with test method 711.1 as modified herein.

4.8.17.1 Applicable to one set.

a. After 711.1.3(d) add the following paragraph:
"711.1.3(d)(1). If any winding insulation resistance is less than 3.11.3.1, the winding shall be dried in accordance with the generator manufacturer's instructions."
b. Paragraph 711.1.3(e) delete: "4 hours" and substitute "8 hours".

4.8.17.2 Applicable to the second set.

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a. Para 711.1.3(b), lines 2 and 3 delete "isolate the armature and field (exciter field only for brushless generators)" and substitute "isolate the armature, main rotor, exciter field and exciter armature in accordance with the generator manufacturer's instructions".
b. 711.1.3(c), add new sentence: "The set shall be connected to an external source of station power; the switchgear and generator resistance heaters shall be energized for six hours immediately prior to and during the test."
c. 711.1.3(d), after the first sentence add: "Immediately after removal from the test chamber, examine the interior of all switchgear cabinets and equipment mounted in the cabinets for surface condensation."
d. 711.1.3(e) delete in its entirety and add: (1) If any winding insulation resistance is less than the minimum recommended by the generator manufacturer, dry out the winding in accordance with manufacturer's instruction. The drying procedure shall start within one half hour after the resistance measurements and shall continue until the minimum insulation resistances required by 3.11.3.1 is obtained. Record the drying time. (2) Within an hour after all insulation resistances are greater than the recommended minimum, reconnect all circuits, disable voltage regulator and start engine and operate as recommended by generator manufacturer. Reconnect the voltage regulator, adjust to rated voltage, and perform test method 608.1.

4.8.18 Preheat test. The preheat test shall be started with engine in the cold condition and run with the set off. The preheat system shall be turned on and operated through at least two heating cycles to ensure coolant is heated properly. Temperature measurement shall be made at each heater inlet to ensure thermostat opens and activates the heaters at 100° F and thermostat closes and deactivates the heaters at 120° F. Voltage and current measurements shall be made at each coolant heater to verify power rating of 5000 watts at each heater. Voltage and current measurements shall be made at the oil heater to verify power rating of 600 watts at the heater. Voltage measurements shall be made to verify that heater circuit is defeated when the set is operating.

4.8.19 Reliability test. The reliability test shall be conducted in accordance with test method 695.1, reliability test, of MIL-STD-705. The test shall be two generators, each run for 875 hours simultaneously. The allowable number of failures shall be two or less. The diesel fuel, during the reliability test, shall conform to VV-F-800, with sulfur content controlled to between 0.55 and 0.65 percent. Both sets shall run 100 hours on JP-5, and the remaining 775 hours on the VV-F-800 fuel specified above. Fuel consumption shall be in accordance with 670.1 of test method 695.1.3.3.1c of MIL-STD-705 for 0/4, 1/4, 2/4, 3/4, and 4/4 of rated load for both diesel and JP-5 fuel. See 6.3.17 for definition of failure.

4.9 Inspection of packaging. Preservation, packing, and marking for each generator set shall be examined for conformance with the quality assurance provisions of MIL-G-28554.

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5. PACKAGING

5.1 Preservation. Preservation of the generator set shall be in accordance with the applicable requirements of MIL-G-28554 for level A, B, or C as specified (see 6.2). Engine cooling systems shall be preserved for all levels using the water and antifreeze procedure of MIL-E-10062.

5.2 Packing. Each preserved generator set shall be packed in accordance with the applicable requirements of MIL-G-28554 for level A, B, or C as specified (see 6.2). For level C packing, any supplemental equipment, disassembled components, parts, etc., shall be consolidated in containers which shall be placed inside the generator housing and secured to prevent movement.

5.3 Marking. The generator sets and any interior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The sets are intended to supply power for multipurpose use in military applications.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the preproduction or first article set to be furnished (see 3.4).
c. When lead acid storage batteries other than as specified are required (see 3.7).
d. When batteries are to be shipped other than as specified (see 3.7).
e. Degree of preservation and of packaging required (see 5.1 and 5.2).
f. When a site requirement module is required (see 3.18).
g. When a remote control station and a remote control cable are required (see 3.17).

6.3 Definitions.

6.3.1 Start. A set is considered to have started when it is operating at rate voltage and speed without the further use of starting aids.

6.3.1.1 Cranking cycle. Each cranking cycle shall consist of 15 seconds of continuous cranking, followed by 30-second rest period, on an inactive set (engine fuel shutoff solenoid de-engerized).

6.3.2 Standby operation. A condition during which a nonoperating set is kept warm and ready to start through use of integral electric winterization kits.

6.3.3 Temperature and humidity damage. Temperature and humidity damage is defined as corrosion, breakage, deformation, reduction of insulation resistance below 50,000 ohms (without benefit of the drying procedure), or conditions

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causing malfunction of any component or part. The minimum suitable insulation resistance for operation with the benefit of the drying procedure is: Minimum Resistance (megohms) =

$$\frac{\text{Machine Voltage} + 1}{1000}$$

The calculated minimum insulation resistances are as follows:

Main Stator = 5.16 Megohms
Main Rotor = 1.25 Megohms
Exciter Stator = 1.2 Meghohms
Exciter Rotor = 1.25 Meghohms

6.3.4 Dangerous flexural vibration. Dangerous flexural vibration is defined as a vibration which occurs at a speed at which maximum stress in the shaft from the flexural vibration exceeds 9,000 psi.

6.3.5 Dangerous torsional critical speed. Dangerous torsional critical speed is defined as the speed at which maximum vibrating stress in the shaft from torsional vibration exceeds 5,000 psi.

6.3.6 Rough handling damage. Rough handling damage is defined as deformation, loosening, breakup, change of fit or any component or part, or any condition resulting in malfunctioning of the set.

6.3.7 Transportation. Transportation is defined as the moving of the set from one location to another by rail, highway, air or water, including the loading and unloading operations required for the relocation.

6.3.7.1 Rail transporation. Rail transportation shall include shocks and stresses from impact at speeds up to and including 5 mph under test conditions specified in MIL-STD-705, method 740.5.

6.3.7.2 Highway transportation. Highway transportation, utilizing commercial, multiaxle trailers, shall include speeds up to and including 40 mph over paved highways, and at speeds up to and including 10 mph over unpaved roads.

6.3.7.3 Water transportation. Water transportation shall include transport by ship or barge, operating over any type of waterway.

6.3.7.4 Air transportation. Air transportation shall include transport by C-130, C-141, or C-5A aircraft.

6.3.8 Weight and dimensions. The weights of the sets are dry weights and do not include the weights of fuel, coolant, lubricating oil, electrolyte, and hydraulic oil.

6.3.9 Dripproof enclosure. A dripproof enclosure is an enclosure so constructed that falling drops of liquid or solid particles striking the

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enclosure at angles from 0 degrees to 15 degrees from the vertical cannot enter the enclosure either directly or by striking and running along a horizontal or inwardly inclinded surface.

6.3.10 Rainproof. An enclosure is considered rainproof when it is so constructed as to prevent any significant accumulation of water within the enclosure and no water has entered the enclosure at a level higher than the lowest live part.

6.3.11 Weathertight. An enclosure is considered weathertight if it is constructed as to minimize the entrance of rain, snow, or airborne particles.

6.3.12 Frequency and voltage regulation. Frequency and voltage regulation is defined as the difference between the no load value and the rated load value divided by the rated load value.

6.3.13 Stable set operating conditions voltage and frequency. Stable set operating conditions are defined as the conditions specified for steady state performance.

6.3.14 Overshoot and undershoot. Reference MIL-STD-705, figure 608.1-VII for definitions of overshoot and undershoot.

6.3.15 Meter accuracy. Accuracy is a number which defines the limit of error expressed as percentage of full scale value. Error is the difference between the indication and the true value of the quantity measured. It is the quantity which, when algebraically subtracted from the indication, gives the true value. A positive error denotes that the indication of the meter is greater than the true value.

6.3.16 Fail-safe. A fail-safe circuit (electrical, mechanical, hydraulic etc.) is one that is designed such that when a failure or malfunction occurs anywhere in the circuit (broken wire, operation of a protective device, mechanical break, etc.), the circuit will return to a condition to restrict the damage to the failed part(s).

6.3.17 Failure. A failure is defined as the inability of an item to perform with specified limits. The Government will identify all failures and will classify them into the following categories.

6.3.17.1 Relevant failure. Any incident in which the item fails to generate electric power within all specified parameter limits as set forth in this specification shall constitute a relevant failure.

6.3.17.2 Nonrelevant faillure. Failures which occur from one of the following causes are considered nonrelevant failures:

a. Failures resulting from operating items beyond requirements as set forth in technical or operator's manuals.

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b. Failures of test instrumentaion or monitoring equipment that is external to the equipment under test.
c. Failures resulting from misuse of equipment or use of equipment outside it's defined capabilities.

6.3.18 Altitude. Altitude is the elevation in feet above mean sea level (msl); the equivalent barometric pressure in millimeters of mercury (mmHg), shown in parenthesis following the altitude [i.e.: 8000 feet (564.4 mmlg)], is based on a barometric pressure of 29.92 inches (760 mm) of mercury under assumed standard msl conditions of 59° F (15° C) and air mass density of 0.001225 grams per cubic centimeter.

6.3.19 Soft load. Soft load is a linear transfer of load from zero to rated or rated to zero in a time not to exceed 20 seconds.

6.3.20 Peaking. Peaking is the addition of generating capacity to the bus to supply additional power for peak load conditions.

6.4 First article model. Any changes or deviations of production sets from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the model will not relieve the contractor of his obligation to furnish sets conforming to this specification.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists and supplier's maintenance and operation manual to be furnished with each engine-generator set.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools, and instructions on shipment of engine-generator sets.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

Custodians:
Army - ME
Navy - YD
Air Force - 11

Preparing activity:
Army - ME

Project 6115-0233

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APPENDIX A

ELECTROMAGNETIC EMISSION AND SUSCEPTIBILITY REQUIREMENTS FOR THE CONTROL OF ELECTROMAGNETIC INTERFERENCE

10. SCOPE

10.1 Scope. This document clarifies the limits, test methods and general procedures to be used for measurement of EMI on Mobile Electric Power Generating Sources (MEPGS). Such clarification is necessary as a result of the recent change to MIL-STD-461. Clarification as to the use of MIL-STD-462 is also included. This document is intended to be used as a clarifying supplement to MIL-STD-461 and MIL-STD-462 and does not replace MIL-STD-461 or MIL-STD-462.

20. APPLICABLE DOCUMENTS

20.1 Issue of documents.

MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference dated 1 April 1980.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of, dated 31 July 1967 and Notice 3 (EL) thereto, dated 9 February 1971.
MIL-HDBK-705	- Generator Sets, Electrical, Measurements and Instrumentations.

30. GENERAL REQUIREMENTS

30.1 Class C equipment requirement. MIL-STD-461 establishes Class C2 for engine generators and associated components, Uninterruptible Power Sets (UPS) and Mobile Electric Power (MEP) equipment supplying power to or used in critical areas. The UM04 limits for conducted emissions, radiated emissions, and radiated emissions susceptibility listed in Part 9 of MIL-STD-461 is applicable to sets rated at 240 kVA or less. The UM04 limits for radiated emissions and radiated susceptibility may be extended to sets with a rating greater than 240 kVA when specified in the procurement document.

30.2 Limits and frequency ranges. Part 9 of MIL-STD-461 establishes the following information:

a. Conducted emissions, 0.015 to 50 MHz, power leads.
b. Radiated emissions, 14 kHz to 1 GHz, electric field.
c. Radiated susceptibility, 2 MHz to 10 GHz, electrical field.

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Drainage provisions shall be 3.3.3 Drainage provisions.

Figures 9-1 and 9-2 of Part 9 establish the criteria for broadband conducted and radiated emissions. The criteria for radiated susceptibility is that the set shall not exhibit any malfunciton, degradation of performance, or indicated in the individual equipment specification when subjected to the following radiated E-fields in the frequency range of 2 MHz to 10 GHz:

FREQUENCY RANGE	E-FIELD (VOLTS/METER)
2 to 400 MHz	10
400 MHz to 10 GHz	5

Above 30 MHz the requirement shall be met for both horizontally and vertically polarized waves.

40. QUALITY CONFORMANCE INSPECTION

40.1 Test procedures. Applicable test procedures for determining conformance to UM04 limits are contained in Notice 3 (EL) to MIL-STD-462. Specifically, Methods CE04, RE02, and RS03 shall be used for conducted emissions, rated emissions, and radiated susceptibility respecitvely. These test procedures are clarified by the attached suplementary test methods.

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TEST METHOD NO. 1

CONDUCTED EMISSIONS

1. Test method. MIL-STD-462, Notice 3 (EL), Method CE04, Procedure (A) Conducted Emissions 50 kHz to 50 MHz, Power Leads.

2. Frequency limits. 0.015 to 50 MHz.

3. Applicability. Applicable in testing to requirement of figure 9.1 of Part 9 of MIL-STD-461.

4. Apparatus.

a. Delete: The use of Line Impedance Stabilization Networks (LISNs) and 50 OHM resistive terminations is not required.
b. Add: Fifty-foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure. With the 50-foot cable connected to the MEPGS output and energized, determine the broadband conducted emissions (0.015 - 50 MHz) on each conductor at the end of the cable at no load, rated kW load at 1.0 pf and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Conducted emissions exceeding the limit of figure 9.1 of MIL-STD-461 shall be cause for rejection.

7. Other:

a. The 50-foot cable shall be energized during the no load test.
b. The 10 microfarad feedthrough capacitors used in superseded test methods are not used in this test method.

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TEST METHOD NO. 2

RADIATED EMISSIONS

1. Test method. Notice 3 (EL), Method RE02, Radiated Emissions, Electrical Field.

2. Frequency limits. 14 kHz to 1 GHz

3. Applicability. Applicable in testing to requirements of figure 9.2 of Part 9 of MIL-STD-461.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The rating of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor.

5. Procedure.

a. In lieu of a continuously welded bond for the counterpoise, bonding straps shall be used.
b. With the 50-foot cable connected to the MEPGS output and energized, determine the radiated emissions (14 kHz - 1 GHz) at no load, rated kW load at 1.0 pf, and rated kW load at 0.8 pf at all voltage and frequency connections.

6. Failure criteria. Radiated emissions exceeding the limit of figure 9.2 of MIL-STD-461 shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no load test.

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TEST METHOD NO. 3

RADIATED SUSCEPTABILITY

1. Test method. Notice 3 (EL), Method RS03, Radiated Susceptibility, 14 kHz to 12.4 GHz, Electrical Field.

2. Frequency limits. 2 MHz to 10 GHz.

3. Applicability. Applicable to testing MEPGS.

4. Apparatus. Add a 50-foot load cable and a resistive/reactive load bank. The ratings of the load bank shall be of sufficient capacity to load the MEPGS to rated capacity at both unity and 0.8 power factor. The output monitor shall be a recording voltage and frequency meter in accordance with MIL-HDBK-705.

5. Procedure.

a. With the 50-foot cable connected to the MEPGS and energized, subject the set to electric fields of 10 V/m (2-400 MHz) and 5 V/m (400 MHz to 10 GHz) at no load, rated kW load at 1.0 PF and rated kW load at 0.8 PF.

b. The set output voltage shall be monitored throughout the test with the recording voltage and frequency meter operating at six inches per hour until voltage or frequency variations as a function of field strength or frequency are noted. Upon observing such variation, the chart speed shall be increased to six inches per minute and the frequency spectrum in question shall be rescanned for acceptance/rejection analysis.

6. Failure criteria. Failure of the output voltage or frequency to remain within the 30-second short-term stability bandwidth (specified in the procurement) when subjected to the specified electrical fields shall be cause for rejection.

7. Other. The 50-foot cable shall be energized during the no load test.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-82058D	Gen Set, Diesel, 750 KW, 50/60 Hertz, Prime Utility

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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